

GeeksforGeeks

PHP Programming Learning Module & Study Guide

Total Topics Covered: **10**

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01

PHP Tutorial

Difficulty Level: Easy

■ Key Technical Concepts

PHP is a popular, open-source scripting language mainly used in web development. It runs on the server side and generates dynamic content that is displayed on a web application. PHP is easy to embed in HTML, and it allows developers to create interactive web pages and handle tasks like database management, form handling, and user authentication. Note: The latest version, PHP 8.4.8, released on 2025, It makes PHP faster, more efficient and adds new features which improving its performance for modern web applications.

■ Code Snippet / Implementation

Not Available

■ Complexity Analysis

Not Available

■ References / Related Links

- PHP Versions (<https://www.geeksforgeeks.org/php/php-versions/>)
- Install PHP in your Windows System (<https://www.geeksforgeeks.org/php/how-to-install-php-in-windows-10/>)
- Set up PHP Project (<https://www.geeksforgeeks.org/php/getting-started-with-php/>)
- Introduction (<https://www.geeksforgeeks.org/php/php-introduction/>)
- PHP Basic Quiz (<https://www.geeksforgeeks.org/quizzes/php-basics-quiz/>)

Source: <https://www.geeksforgeeks.org/php-tutorial/>

02

PHP Variables

Difficulty Level: Not Available

■ Key Technical Concepts

A variable in PHP is a container used to store data such as numbers, strings, arrays, or objects. The value stored in a variable can be changed or updated during the execution of the script. To declare a variable in PHP, you simply assign a value to it using the \$ symbol followed by the variable name. PHP variables are case-sensitive and must start with a letter or an underscore, followed by any number of letters, numbers, or underscores.

■ Code Snippet / Implementation

```
<?php$name="XYZ";// String$age=30;// Integer$salary=45000.50;// Float$isEmployed=true;// Boolean?>
```

■ Complexity Analysis

Not Available

■ References / Related Links

Not Available

Source: <https://www.geeksforgeeks.org/php-variables/>

03

PHP Data Types

Difficulty Level: Basic

■ Key Technical Concepts

In PHP, data types define the kind of value a variable can hold. PHP is a loosely typed language, meaning you don't need to declare the data type of a variable. It is automatically assigned based on the value. But it is important to understand data types because it is important for writing reliable, bug-free, and efficient code. PHP supports 8 main data types, grouped into three categories:

■ Code Snippet / Implementation

```
<?php$age=25;echo$age;// Output: 25?>
```

■ Complexity Analysis

Not Available

■ References / Related Links

- PHP Strings (<https://www.geeksforgeeks.org/php/php-strings/>)
- PHP Arrays (<https://www.geeksforgeeks.org/php/php-arrays/>)
- PHP Objects (<https://www.geeksforgeeks.org/php/php-objects/>)
- var_dump() (https://www.geeksforgeeks.org/php/php-var_dump-function/)
- is_int(): (https://www.geeksforgeeks.org/php/php-is_int/)

Source: <https://www.geeksforgeeks.org/php-data-types/>

04

PHP Arrays

Difficulty Level: Not Available

■ Key Technical Concepts

Arrays are one of the most important data structures in PHP. They allow you to store multiple values in a single variable. PHP arrays can hold values of different types, such as strings, numbers, or even other arrays. Understanding how to use arrays in PHP is important for working with data efficiently. Indexed arrays use numeric indexes starting from 0. These arrays are ideal when you need to store a list of items where the order matters.

■ Code Snippet / Implementation

```
<?php$fruits=array("apple","banana","cherry");echo$fruits[0];// Outputs: apple?>
```

■ Complexity Analysis

Not Available

■ References / Related Links

- Indexed arrays (<https://www.geeksforgeeks.org/php/php-indexed-arrays/>)
- Associative arrays (<https://www.geeksforgeeks.org/php/associative-arrays-in-php/>)
- Multidimensional arrays (<https://www.geeksforgeeks.org/php/multidimensional-arrays-in-php/>)
- PHP Array Function (<https://www.geeksforgeeks.org/php/php-array-functions-complete-reference/>)
- PHP Array Iteration (<https://www.geeksforgeeks.org/php/how-to-iterating-through-an-array-in-php/>)

Source: <https://www.geeksforgeeks.org/php-arrays/>

05

PHP | Functions

Difficulty Level: Not Available

■ Key Technical Concepts

A function in PHP is a self-contained block of code that performs a specific task. It can accept inputs (parameters), execute a set of statements, and optionally return a value. Once a function is declared, it can be invoked (called) by simply using the function name followed by parentheses.

■ Code Snippet / Implementation

```
<?php// Creating a function that adds two numbersfunctionaddNumbers($a,$b){// code to be executed$result=$a+$b;// returning the sumreturn$result;}// Calling the functionechoaddNumbers(5,10);?>
```

■ Complexity Analysis

Not Available

■ References / Related Links

- PHP String Functions (<https://www.geeksforgeeks.org/php/php-string-functions/>)
- PHP Array Functions (<https://www.geeksforgeeks.org/php/php-array-functions-complete-reference/>)
- PHP Date and Time Functions (<https://www.geeksforgeeks.org/php/php-date-time/>)

Source: <https://www.geeksforgeeks.org/php-functions/>

06

PHP Strings

Difficulty Level: Not Available

■ Key Technical Concepts

In PHP, strings are one of the most commonly used data types. A string is a sequence of characters used to represent text, such as words and sentences. Strings are enclosed in either single quotes (' ') or double quotes (" "). There are four different ways to define a string literal in PHP:

■ Code Snippet / Implementation

```
<?php$site='Welcome to GeeksforGeeks';echo$site;?>
```

■ Complexity Analysis

Not Available

■ References / Related Links

- `strlen()` (<https://www.geeksforgeeks.org/php/php-strlen-function/>)
- `strrev()` (<https://www.geeksforgeeks.org/php/php-strrev-function/>)
- `strtoupper()` (<https://www.geeksforgeeks.org/php/php-strtoupper-function/>)
- `strtolower()` (<https://www.geeksforgeeks.org/php/php-strtolower-function/>)
- `strpos()` (<https://www.geeksforgeeks.org/php/php-strpos-strcmp-functions/>)

Source: <https://www.geeksforgeeks.org/php-strings/>

07

PHP Loops

Difficulty Level: Not Available

■ Key Technical Concepts

In PHP, Loops are used to repeat a block of code multiple times based on a given condition. PHP provides several types of loops to handle different scenarios, including while loops, for loops, do...while loops, and foreach loops. In this article, we will discuss the different types of loops in PHP, their syntax, and examples.

■ Code Snippet / Implementation

```
for ( Initialization; Condition; Increment/Decrement ) { // Code to be executed }
```

■ Complexity Analysis

Not Available

■ References / Related Links

Not Available

Source: <https://www.geeksforgeeks.org/php-loops/>

08

Not Available

Difficulty Level: Not Available

■ Key Technical Concepts

Your All-in-One Learning Portal. It contains well written, well thought and well explained computer science and programming articles, quizzes and practice/competitive programming/company interview Questions.

■ Code Snippet / Implementation

Not Available

■ Complexity Analysis

Not Available

■ References / Related Links

Not Available

Source: <https://www.geeksforgeeks.org/php-conditional-statements/>

09

Not Available

Difficulty Level: Not Available

■ Key Technical Concepts

Your All-in-One Learning Portal. It contains well written, well thought and well explained computer science and programming articles, quizzes and practice/competitive programming/company interview Questions.

■ Code Snippet / Implementation

Not Available

■ Complexity Analysis

Not Available

■ References / Related Links

Not Available

Source: <https://www.geeksforgeeks.org/php-form-handling/>

10

PHP | Sessions

Difficulty Level: Not Available

■ Key Technical Concepts

Note: PHP sessions allow different PHP pages to share data. By calling `session_start()` on every page, the same session data becomes accessible across all pages, effectively connecting them during a user's visit. Using PHP sessions involves several key steps: starting a session, storing data in session variables, retrieving data, and eventually destroying the session when no longer needed.

■ Code Snippet / Implementation

```
<?phpsession_start(); // Start the session?>
```

■ Complexity Analysis

Not Available

■ References / Related Links

- PHP Cookies (<https://www.geeksforgeeks.org/php/php-cookies/>)

Source: <https://www.geeksforgeeks.org/php-sessions/>